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Alkylation of electron-deficient olefins through conjugate addition of organozinc reagents: An update

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Abstract: This review focuses on the applications of organozinc reagents in conjugate addition reactions. Organozinc compounds are a class of organometallic reagent used in various reactions like addition, substitution and coupling reactions. The conjugate addition or 1, 4-addition is one of the most useful bond forming strategies used in organic synthesis since broad range of nucleophiles can be employed in the reaction. Addition of organozinc to electron-deficient olefins attracts synthetic chemists as it preferably undergoes conjugate addition over 1,2-addition unlike other organometallic reagents, such as Grignard reagent or organolithium. Many natural products were also synthesized by applying the conjugate addition reaction of organozinc. Moreover, organozinc reagents have superior functional group compatibility and can be easily prepared relative to other organometallic compounds. Consequently, conjugate additions of organozinc reagents were highly studied and having a review on it is very important at present. Though a few reviews are available, they are either not directly on the organozinc reagents or on the conjugate addition. We hope that this review article will be of great use to organic researchers.

1. Introduction

Organozinc reagents are known to be the first organometallic compounds ever synthesized in organic chemistry having a C–Zn chemical bond.^[1] It was first synthesized by Edward Frankland in the 1850s. As carbon and zinc atoms have electronegativity differences, the C–Zn bond is always polarized towards carbon. Hence, organozinc reagents are used as nucleophiles in organic reactions, such as addition reactions, substitution reactions, coupling reactions, etc.^[2] However, initially when they were employed as nucleophiles in organic synthesis, poor yields of the corresponding products were formed.^[1b] As a result, synthetic chemists preferred more reactive Grignard reagents over organozinc reagents as nucleophiles. Consequently, organozinc reagents were overlooked for more than 10 decades until the 1970s when Negishi et al.^[2c,3] noticed that use of a palladium or nickel catalyst increases the reactivity of organozinc reagent. Today organozinc reagents are one of the most useful classes of organometallic compounds because of their advantages, such as ease of preparation and superior functional group compatibility relative to Grignard reagents and organolithium (Figure 1). Moreover, their brilliant reactivity in the presence of a catalyst makes them a very powerful tool for natural product

synthesis. Organozinc reagents can be divided into three types, such as organic zinc halogen compound (R–Zn–X), dialkyl or diarylzinc (R–Zn–R), lithium or magnesium zincate. However, it is observed that different methods of preparation of organozinc reagents could lead to dissimilar structures showing their special compatibilities. Organozinc reagents undergo conjugate addition or 1,4-addition like organocopper reagents^[4] when react with α,β -unsaturated carbonyl compounds.

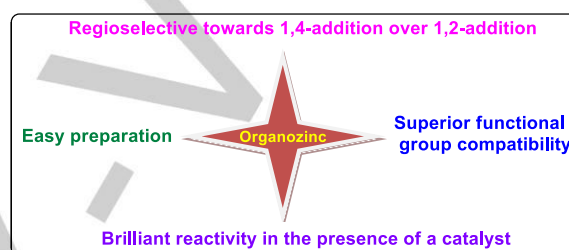


Figure 1. Advantages of using organozinc reagents

Conjugate addition reaction is the addition of a nucleophile to an electron-deficient olefins. A carbanion is generated first in the reaction, which can either be protonated to give the β -substituted product or an electrophilic species can be added to produce the α,β -disubstituted product. The conjugate addition is one of the most useful bond forming strategies and has been broadly used in organic synthesis.^[5] The reason of versatility of the conjugate addition is mainly because of the different types of nucleophiles (organometallic reagents, Michael donors, other carbanions) that can be employed in the reaction.^[6]

Dialkylzinc reagents were first developed for conjugate additions through catalytic process. Thereafter, different types of organozinc reagents were used in conjugate addition. There are many natural products which were synthesized by the conjugate addition of dialkylzinc. A few such natural products are depicted in Figure 2. For instance, R-(–) muscone,^[7] clavularin B,^[8] pumiliotoxin C,^[9] and PEG₁ methyl ester^[10] were synthesized by applying the conjugate addition of organozinc reagent. For that reason, many research articles were published on the conjugate addition of organozinc. A few reviews and book chapter were also published, but they are much old.^[11] More recently, a few review articles have been published, but they are either not specific to organozinc reagents or do not focus on conjugate addition reactions.^[12] Therefore, we planned to write a review primarily on the conjugate addition of organozinc reagents. The main objectives of this article are (i) to highlight the important research on conjugate additions of organozinc reagents, (ii) to highlight the limitations and possibilities for improvement, (iii) to attract the synthetic chemists so that they can use the strategy in the synthesis of natural products and drug molecules. We have discussed here the reactions and their results since 2005 to till date. We have divided the article into two major sections

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viz non-asymmetric conjugate addition and asymmetric conjugate addition and subdivided the latter section into copper-catalyzed and copper-free asymmetric conjugate addition.

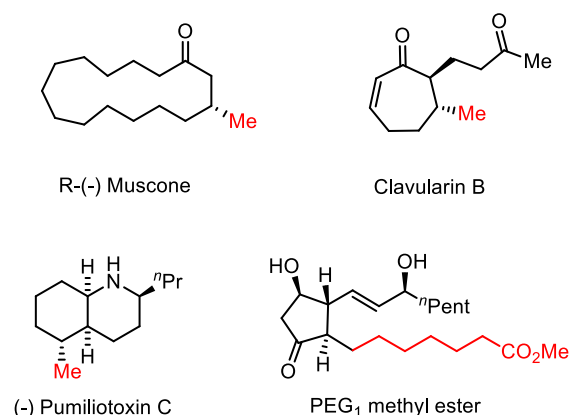
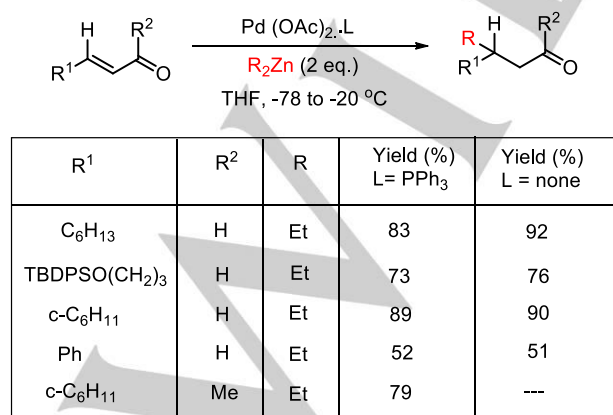


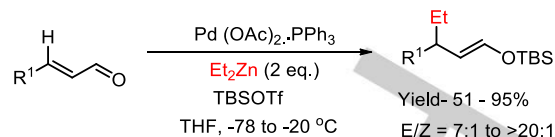
Figure 2. Natural products synthesized *via* conjugate addition of dialkylzinc

2. Non-asymmetric conjugate addition

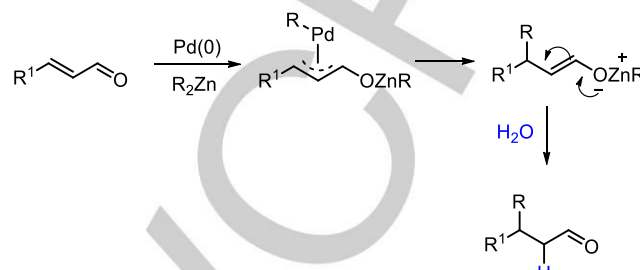
In non-asymmetric conjugate addition the product has no chirality, though new stereogenic centre(s) may generate. One such conjugate addition of organozinc reagents to conjugated enals was reported using Pd catalyst without competing 1,2-additions. The reaction produced good to excellent yields in the presence of 5 mol% of Pd(OAc)₂.PPh₃ precatalyst in THF solution at -78 to -20 °C (Scheme 1).^[13] Without using the PPh₃ ligand, the reaction still gave comparable yield of 1,4-adducts (Scheme 1). However, difficulties in the purification of products owing to the precipitation of persistent palladium by-products encourage the authors to use the ligand. The intermediate zinc enolates can be trapped by quenching the reaction with TBSOTf which afforded predominantly the (*E*)-enol silanes with moderate to excellent yield (Scheme 2). However, the 1,4-additions to conjugated acyl oxazolidinones, nitriles or esters were unsuccessful. A plausible mechanistic pathway is delineated in Scheme 3.



Scheme 1. Conjugate addition of dialkylzinc to conjugated enals

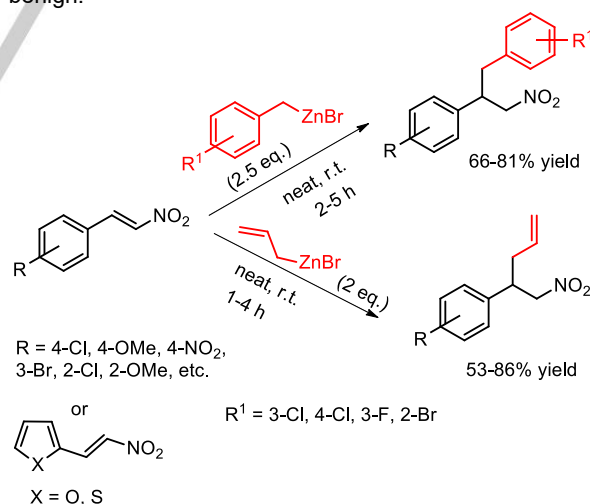


Scheme 2. Trapping of intermediate zinc enolates



Scheme 3. The plausible mechanism for Pd-catalyzed conjugate addition of dialkylzinc

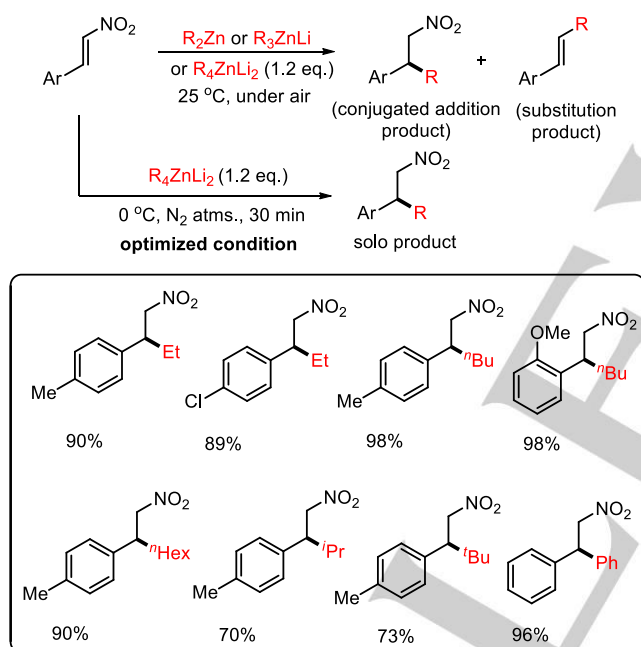
Conjugate addition of allylzinc and benzylzinc reagents to β-nitroalkenes was reported by Wang and co-workers. The reactions were performed without solvent and catalyst at room temperature which produced the desired products in moderate to good yield (Scheme 4).^[14] It was noticed that the presence of electron withdrawing groups (EWGs) on the phenyl ring of nitrostyrenes decreased the product yield. In contrast, electron rich heterocyclic nitroalkenes furnished higher yield. With benzylzinc bromide the reaction gave lower yield and required comparatively longer time. When the reaction of benzylzinc bromide and β-nitroalkenes were performed in THF solvent, no 1,4-addition product was formed which indicated the necessity of solvent-free reaction. This reaction has a few advantages, such as milder conditions, shorter reaction time and environmentally benign.



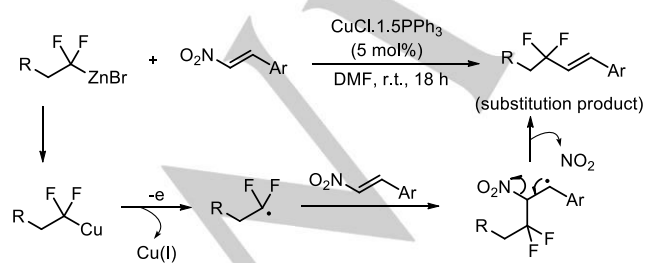
Scheme 4. Conjugate addition of allylzinc and benzylzinc reagents to β-nitroalkenes

Recently, Aera et al. disclosed the addition of lithium tetraorganozincates to nitroolefins (Scheme 5).^[15] They initially investigated the reaction of diethylzinc or lithium zincates (Et₃ZnLi and Et₄ZnLi₂) with β-nitrostyrene using deep eutectic

solvent, a mixture of choline chloride and D-(-) fructose at 25 °C under air. Although a good amount of conjugated addition product was obtained, a substitution product was also formed (8:2 to 9:1) as by-product in which the nitro group was displaced by the nucleophilic alkyl group of organozinc reagents. When the reaction was performed at 0 °C under nitrogen atmosphere with lithium tetraorganozincate (Et_4ZnLi_2), 98% of the desired conjugated addition product was obtained as the sole product. It was observed that the strict exclusion of air or lowering the temperature to 0 °C was crucial to suppress or completely remove the substitution product in the reaction. Lithium tetraorganozincate was prepared by mixing 2 equivalent of RLi (0.5 M benzene/cyclohexane) with R_2Zn . A variety of tetraorganozincates such as Et_4ZnLi_2 , $^n\text{Bu}_4\text{ZnLi}_2$, $n\text{-Hexyl}_4\text{ZnLi}_2$, $^i\text{Pr}_4\text{ZnLi}_2$ and $^t\text{Bu}_4\text{ZnLi}_2$ were employed which produced good to excellent yields. Dilman and co-workers reported a reaction of *gem*-difluorinated organozinc reagent with β -nitrostyrene in the presence of Cu(I) catalyst. Surprisingly, the reaction did not give any conjugated addition product and produced only substitution product instead (Scheme 6).^[16]



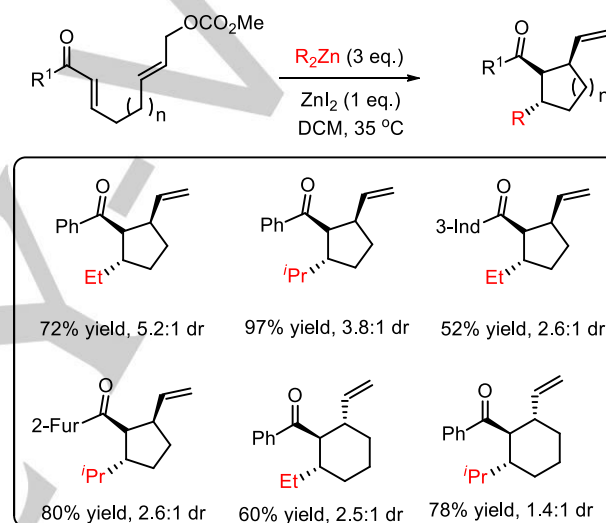
Scheme 5. Conjugate addition of tetraorganozincates to nitroolefins



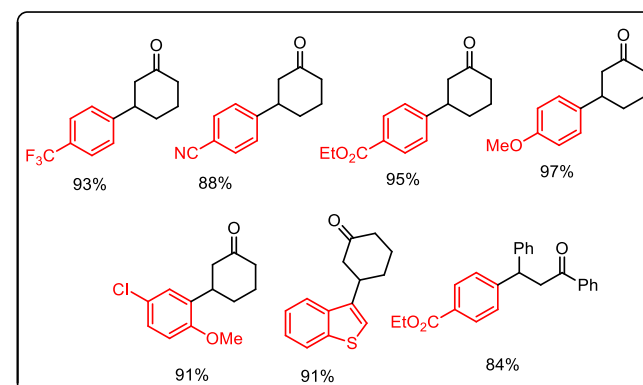
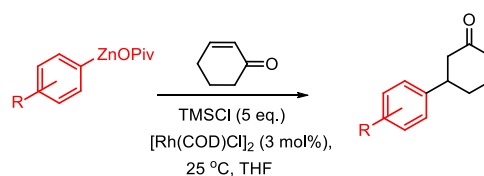
Scheme 6. Substitution reaction of organozinc over conjugate addition

An interesting conjugate addition/cyclization was developed by Krische et al. in which enones possessing

pendant allylic carbonates were treated with dialkylzinc. The reaction was promoted by ZnI_2 (1 eq.) and performed in DCM solvent at 35 °C which afforded 5- and 6-membered ring products through tandem conjugate addition-cycloallylation in good to excellent yield (Scheme 7).^[17] The reaction generated three new stereogenic centers. The diastereoselectivity was found up to 5.3:1. The reaction first produced intermediate zinc enolates which underwent $\text{S}_\text{N}2'$ type substitution reactions to appendant allylic carbonates in the absence of a copper or palladium catalyst. Heteroaromatic enones containing 3-indolyl and 2-furyl moieties also successfully produced the desired products. The use of organozinc reagent $^i\text{Pr}_2\text{Zn}$ provided higher yield of the products than Et_2Zn because of more reactivity of former towards conjugate addition. However, the use of different iodide ion, such as Bu_4NI reduced the product yield.



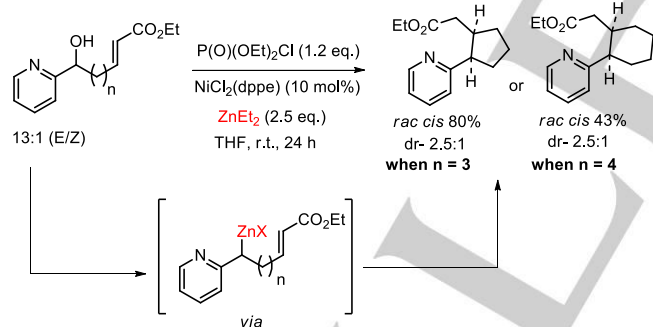
Scheme 7. Conjugate addition/cyclization of dialkylzinc to enones



Scheme 8. Conjugate addition of arylzinc pivalates to cyclohex-2-enone

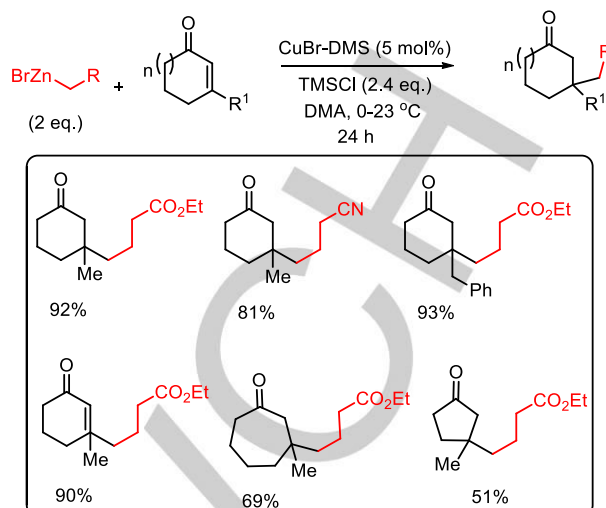
The conjugate addition of arylzinc pivalates to cyclohex-2-enone in the presence of 3 mol% of $[\text{Rh}(\text{COD})\text{Cl}]_2$ and TMSCl (5 eq.) in THF solvent at 25 °C was developed (Scheme 8).^[18] The products were obtained in 88–97% yield. Arylzinc pivalates were first prepared by either magnesium insertion or Halogen/Mg exchange followed by transmetalation with $\text{Zn}(\text{OPiv})_2$. Heterocyclic zinc reagent also underwent 1,4-addition to give the corresponding product with excellent yield. Arylzinc pivalates also reacted with acyclic Michael acceptor (*E*)-chalcone giving very good yield of product. When used α,β -unsaturated esters, decomposition of starting materials occurred and only low conversions into the desired products was observed by GC-analysis.

Jarvo and co-workers reported an intramolecular conjugate addition of *in situ* generated organozinc compound from the unsaturated ester of 2-(hydroxymethyl)pyridines to synthesize pyridine-2-cyclopentane or -cyclohexane (Scheme 9).^[19] The reaction was performed by using diethylchlorophosphate, nickel catalyst and diethylzinc. The reaction proceeded through hydrogenolysis of 2-(hydroxymethyl)pyridines. The function of diethylchlorophosphate was to activate the alcohol and diethylzinc was used as transmetalating agent. Without nickel catalyst or diethylchlorophosphate no product formation was observed. In the proposed mechanism a benzylzinc reagent bearing a pendant α,β -unsaturated ester was first formed which underwent smooth cyclization to produce the desired cyclopentane or cyclohexane in moderate to good yield with good diastereoselectivity. Attempts to construct 3- or 7-membered rings were unsuccessful and resulted primarily in hydrogenolysis.



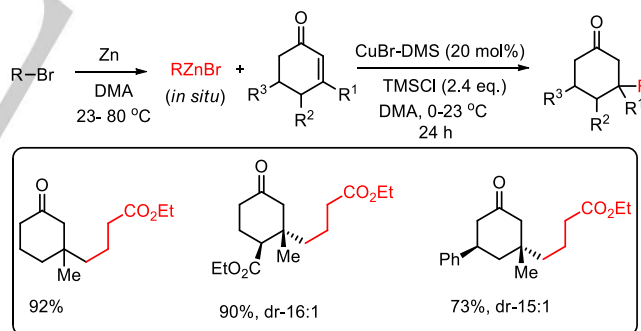
Scheme 9. Intramolecular conjugate addition of *in situ* generated organozinc reagent

Monoorganozinc compounds exhibit excellent reactivity towards β,β -disubstituted enones in the presence of TMSCl and $\text{CuBr}\cdot\text{DMS}$ in polar solvent *N,N*-dimethylacetamide (DMA) at 0 to 23 °C to synthesize various cyclic functionalized β -quaternary ketones (Scheme 10).^[20] The reaction can furnish five-, six-, and seven-membered ring products in high yields. Monoorganozinc compounds can readily be obtained from alkyl bromides which participate in conjugate addition.



Scheme 10. Conjugate addition of organozinc reagents to synthesize β -quaternary ketones

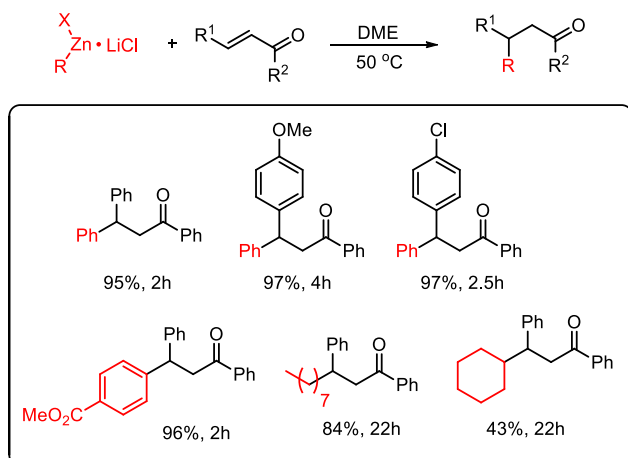
The reaction was also attempted in one-pot process by generating the organozinc reagent *in situ* giving excellent yields of β -quaternary ketones. Cyclohexenone containing γ - and δ -substitution also produced high yields of desired products with excellent diastereoselectivities (Scheme 11). Stabilized monoorganozinc reagents, such as enolate, homoenolate and allylzinc bromide did not respond to the addition reaction. In one-pot procedure 20 mol% of catalyst was required while 5 mol% of catalyst was needed for the direct reaction of organozinc and cyclohexenone.



Scheme 11. One-pot conjugate addition of organozinc reagents

A catalyst-free conjugate addition of aryl and alkylzinc halides to nonenolizable unsaturated ketones was developed (Scheme 12).^[21] The organozinc halides were prepared from the direct insertion of zinc into organic halides in the presence of LiCl . The reaction was carried out in 1,2-dimethoxyethane (DME) solvent at 50 °C which gave excellent yield of β,β -disubstituted products. A broad range of both alkyl and arylzinc halides and ketones were successfully utilized in the reactions. Two different ether solvents THF and DME were studied for the reaction and found from the DFT calculations that due to presence of two oxygen atoms in DME, it forms chelate

compound with the lithium cation present in the transition state and plays a key role in lowering the activation energy.

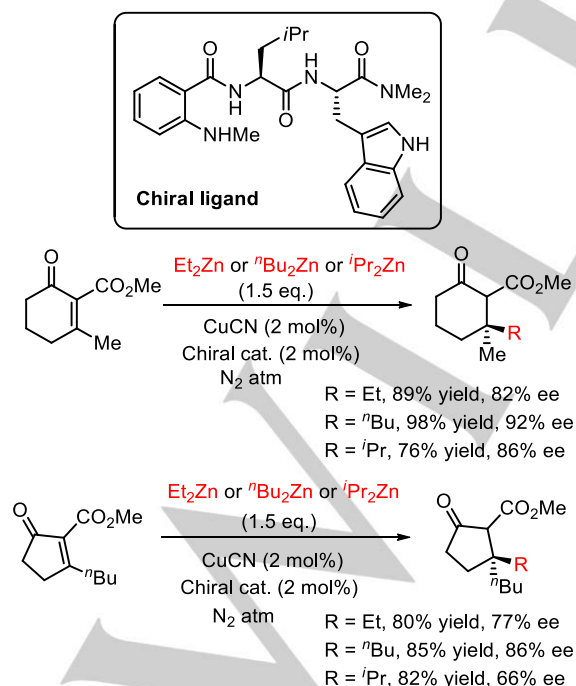


Scheme 12. A catalyst-free conjugate addition of aryl and alkylzinc halides

3. Asymmetric conjugate addition

3.1. Copper-catalyzed asymmetric conjugate addition

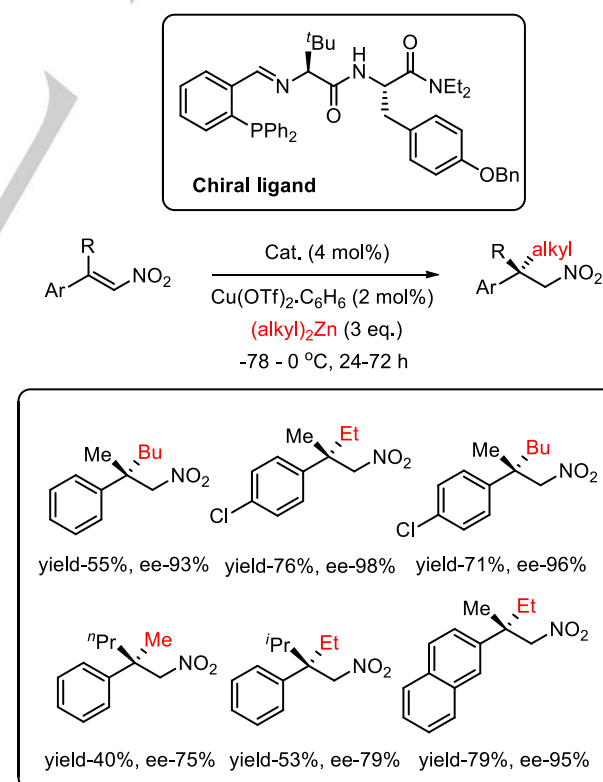
In 2002, an improved method for copper-catalyzed conjugate addition of organozinc to electron-deficient olefins was developed by Alexakis et al. in which >99% of enantiomeric excess was observed.^[22] In 2005, another asymmetric conjugate addition of dialkylzinc to cyclic enones was disclosed by Hoveyda and co-workers. Tetrasubstituted alkene moiety was considered for the reaction and the product obtained was having two quaternary stereogenic centers (Scheme 13).^[23] The



Scheme 13. Asymmetric conjugate addition of dialkylzinc to cyclic enones

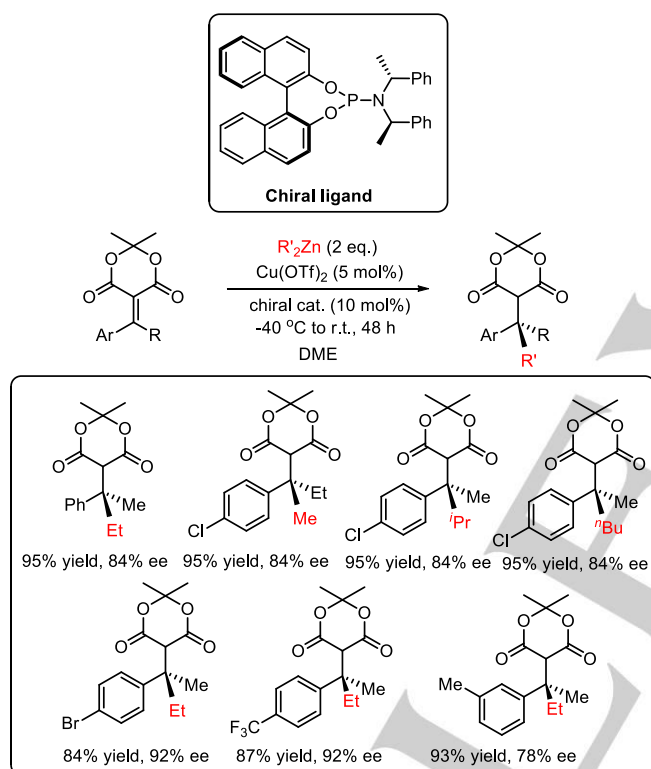
reaction was catalyzed by air-stable CuCN along with chiral ligand that contains an anthranilic acid-based N-terminus. The reaction produced up to 98% conversion with 95% enantiomeric excess. Both six- and five-membered cyclic enones were utilized in the reaction. All the reactions were conducted in undistilled toluene solvent under nitrogen atmosphere at -15 to 0 °C. Interestingly, purified solvent offered lower enantioselectivities. Different commercially available dialkylzinc reagents were used in the reaction. This was the first enantioselective conjugate addition of any alkylmetal reagent to tetrasubstituted olefins. One important review on copper-catalyzed enantioselective conjugate addition was published in 2008 by Alexakis et al.^[24]

In the same year, the same group developed another asymmetric conjugate addition of dialkylzinc reagents to acyclic nitroalkenes resulting to the formation of all-carbon quaternary carbon stereogenic centers (Scheme 14).^[25] Enantiopure organic compounds containing quaternary carbon are important because of their relevance to the synthesis of biologically active compounds. Reactions were performed in the presence of Cu(OTf)₂ catalyst in toluene at -78 to 0 °C which produced products with high enantiomeric excess (up to 98%). A variety of dialkylzinc reagents and nitroolefins were used in the reactions. Chiral peptide based ligand was used to induce chirality in the product. The enantiopure nitroalkanes may be converted effectively to synthetically versatile enantiopure compounds, such as the derived nitriles, amines, carboxylic acids, and aldoximes.

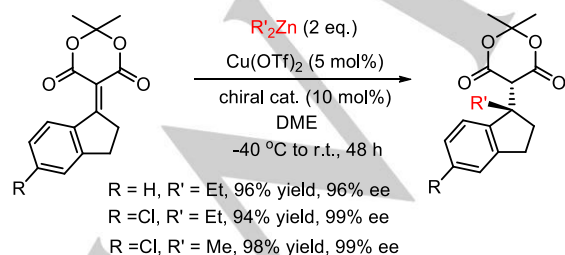


Scheme 14. Asymmetric conjugate addition of dialkylzinc to acyclic nitroalkenes

Another conjugate addition of dialkylzinc to 5-(1-arylalkylidene) Meldrum's acids was developed by Fillion et al. The products contain all-carbon benzylic quaternary stereocenters (Scheme 15).^[26] The reaction was catalyzed by Cu(II) salt along with commercially available chiral ligand. The reactions were carried out in 1,2-dimethoxyethane (DME) solvent at -40 °C to room temperature for 48 h. In most of the cases reactions produced excellent yield with very good enantioselectivities. 5-(1-Substitutedalkylidene) Meldrum's acids were efficiently prepared through Knoevenagel condensation of Meldrum's acid with ketones.^[27] Different alkyl substitutions on the alkene acceptor did not affect the yield of the reactions. The conjugate addition of dialkylzinc was then extended to 5-(dihydroindenylidene) Meldrum's acids which exhibited excellent enantioselectivities (Scheme 16).

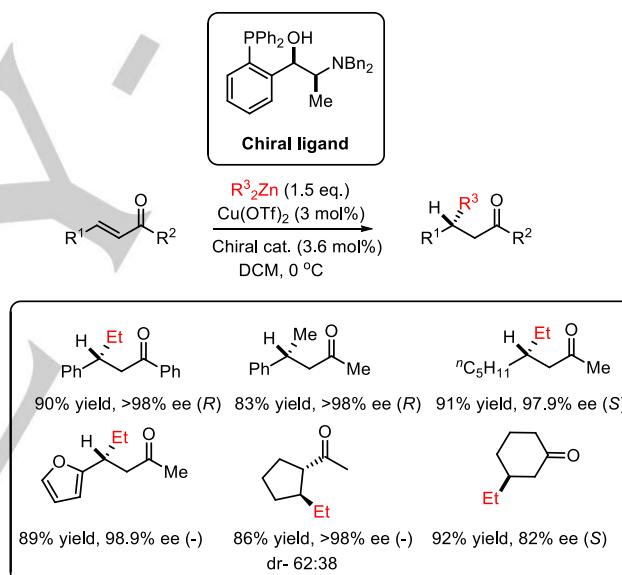


Scheme 15. Asymmetric conjugate addition of dialkylzinc to 5-(1-arylalkylidene) Meldrum's acids



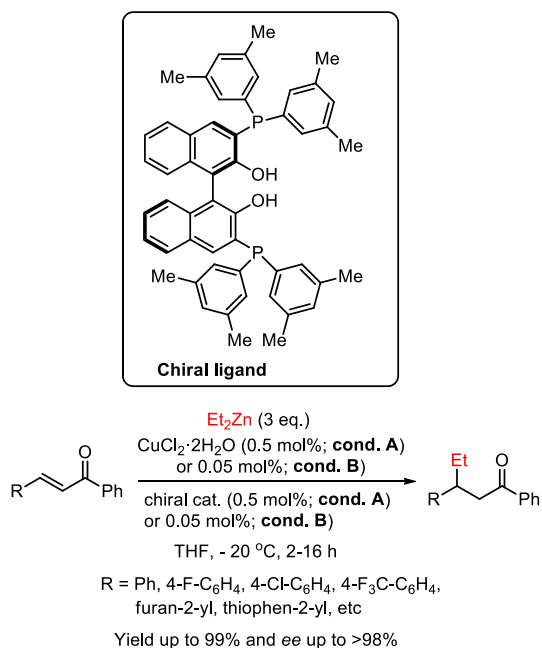
Scheme 16. Asymmetric conjugate addition of dialkylzinc to 5-(dihydroindenylidene) Meldrum's acids

Nakamura and co-workers developed an alanine-derived aminohydroxyphosphine ligand for the asymmetric conjugate addition of dialkylzinc reagents to α,β -unsaturated carbonyl compounds catalyzed by Cu(OTf)₂ (Scheme 17).^[28] The reaction was performed in DCM solvent at 0 °C. A variety of open chain and cyclic enones were used as substrates which produced very good yield of addition products. The asymmetric catalyst which was a newly developed tridentate ligand, induced high enantioselectivity in the products. The ligand was synthesized in one step by treating lithiated triphenylphosphine to *N,N*-dibenzylalanine. It was observed that choice of solvent for the reaction was crucial. Noncoordinating solvents were found superior to coordinating solvents. It was also noticed that ligands derived from bulkier aminoaldehydes offered lower selectivity. Mechanistic investigations suggested that the reaction proceeded *via* a highly ordered transition state by the coordination of phosphorus and nitrogen atoms of the ligand to the Cu(III) and Zn(II) atoms respectively and of the oxygen anion to both the metal centers.



Scheme 17. Asymmetric conjugate addition of dialkylzinc to α,β -unsaturated carbonyls

A similar asymmetric conjugate addition of organozinc reagents to enones was reported using binol derivatives as asymmetric ligand and catalyzed by CuCl₂·2H₂O (Scheme 18).^[29] Two reaction conditions were employed, in one condition 0.5 mol% of copper catalyst and ligand were used. In the other condition 0.05 mol% of catalyst and ligand were used. 3 Eq. of organozinc were used in both the conditions. Both the conditions showed excellent results. For instance, when R = Ph, 91% yield and more than 98% ee of (*S*) stereoisomer was observed in condition A, while 99% of yield and 96% ee of (*S*) isomer was obtained in condition B. Heteroarylenones also exhibited very good results. It was believed that the asymmetric ligand coordinated to multiple metal centers provided the effective chiral catalysis.



Scheme 18. Asymmetric conjugate addition of dialkylzinc to enones in the presence of binol derivative as chiral ligand.

In the same year, a similar asymmetric conjugate addition of dialkylzinc to enone was reported by Wang and co-workers in which a chiral diphosphite ligand synthesized from the reaction of D-mannitol and chlorophosphoric acid diary ester was used for asymmetric induction (Figure 3).^[30] The reaction was catalyzed by $\text{Cu}(\text{OTf})_2$ and carried out in toluene solvent at -20 °C. The reaction gave high enantioselectivity (up to 93%) for cyclic product (cyclohexanone and cyclopentanone), however, open chain product exhibited poor enantioselectivity (up to 53%). Enantioselectivity was mainly determined by the configuration of the diaryl phosphite moieties.

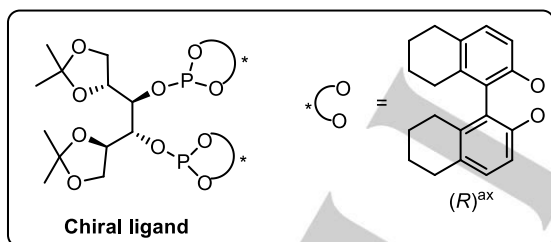


Figure 3. Asymmetric ligand used in the conjugate addition of dialkylzinc to enone

Copper-catalyzed asymmetric conjugate addition of organozinc reagents to enones using functionalized BINOL-mono-PHOS was developed by Endo and Shibata et al. (Figure 4a).^[31] The reaction was performed in the presence of $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ (5 mol%) catalyst along with the asymmetric ligand (10 mol%) in THF solvent at -20 °C. Diethyl- and dimethylzinc were used (3 eq.) as the nucleophilic partner. The yield and enantioselectivity were significantly improved when a bulky hydroxy group was incorporated at the 3'-position of BINOL-mono-PHOS. The reaction was equally effective in both acyclic

enones and cyclohexenone. Up to 97% of yield and 98% of enantioselectivity was observed for the products. They again reported a copper catalyzed asymmetric conjugate addition of dialkylzinc reagents to enones using another asymmetric ligand, functionalized BINAM-PHOS (Figure 4b).^[32] The reaction was carried out by reacting enones with dialkylzinc (3 eq.) in the presence of $\text{Cu}(\text{acac})_2$ (2.5 mol%) and asymmetric ligand (5 mol%). Excellent yield and enantioselectivity were observed in all the cases. The incorporation of benzoyl derivatives on nitrogen atoms was crucial for obtaining high enantioselectivity. It is worth to mention that BINAM-PHOS derivatives offered the opposite major enantiomers to that achieved using BINOL-mono-PHOS derivatives.

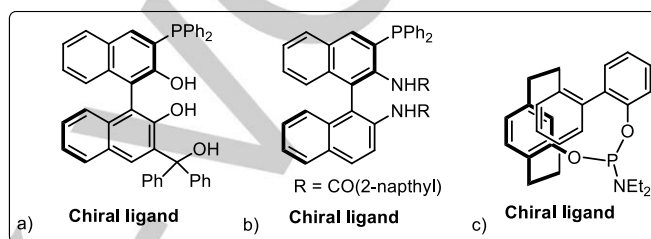
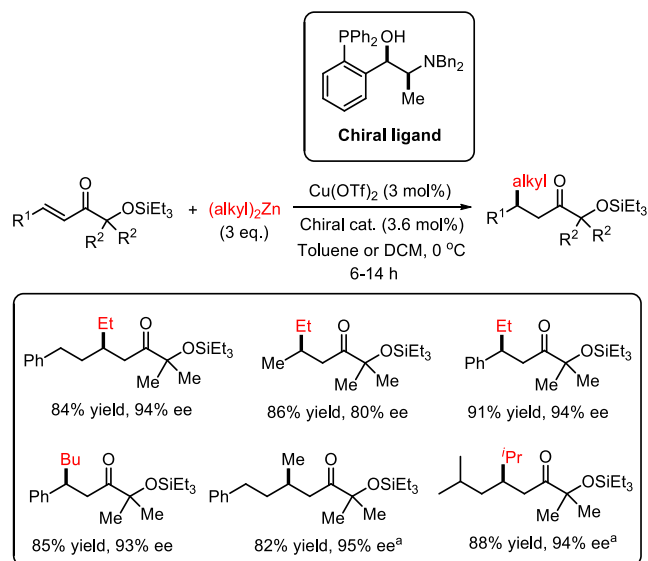


Figure 4. Asymmetric ligands used in the conjugate addition of dialkylzinc to enones

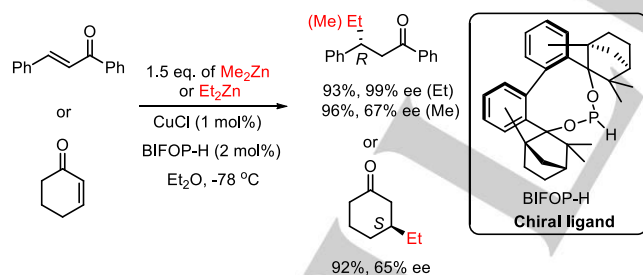
Another copper-catalyzed asymmetric conjugate addition of diethylzinc to enones was developed by Jiang and co-workers using [2.2]paracyclophane-derived monodentate phosphoramidite as asymmetric ligand (Figure 4c).^[33] The reaction required $\text{Cu}(\text{OTf})_2$ (1 mol%) as catalyst and 1.2 mol% of ligand, whereby 1.5 eq. of diethylzinc was employed in the reaction. Reactions were performed at 0 °C in hexane/toluene. Excellent yield (up to 98%) and enantioselectivity (up to 95%) of products were obtained. The asymmetric ligand was derived from a flexible 4,12-disubstituted [2.2]paracyclophane. When aryl group attached to alkene moiety contains an *ortho* substitution, enantioselectivity reduced greatly. When aryl group attached to carbonyl moiety contains electron withdrawing *para* substitution, enantioselectivity was relatively lower.

A research group from Spain developed a copper catalyzed asymmetric conjugate addition of organozinc reagents to α' -oxy enones. Dialkylzinc was reacted with α' -oxy enones in the presence of $\text{Cu}(\text{OTf})_2$ along with a asymmetric catalyst in toluene or DCM at 0 °C (Scheme 19).^[34] The asymmetric ligand used in the reaction was alanine-derived aminohydroxyphosphine, which was developed by Nakamura et al. (Scheme 17). The reaction gave high yield with excellent enantioselectivity of desired products. Different dialkylzinc reagents were employed in the reaction which furnished excellent enantioselectivity. It is worth to mention that the flanking quaternary carbon center was crucial in the reaction for chiral induction which was indicated by the relatively poor enantioselectivity of products in which flanked quaternary carbon changed to secondary carbon center.



Scheme 19. Asymmetric conjugate addition of dialkylzinc to α' -oxy enones.
^aReaction performed in DCM solvent.

Recently, an enantioselective Cu(I) or Cu(II) catalyzed conjugate addition of dialkylzinc to enones and cyclohexenone were reported using ferrocenyl-based phosphorus ligands, e.g. biphenyl-2,2'-bisferrocenyl hydrido phosphate (BIFOP-H) (Scheme 20).^[35] The 1,4-addition of Et_2Zn to enones catalyzed by CuCl•BIFOP-H yields up to 93% and 99% ee. Interestingly, CuCl gave better enantioselectivity in 1,4-additions to enones (CuCl: 76% ee; Cu(OTf)₂: 49% ee; CuCl₂: 42% ee) while Cu(OTf)₂ was found better in 1,4-additions to cyclohexenone (Cu(OTf)₂: 65% ee; CuCl: 20% ee). However, when chromone was employed as the Michael acceptor in the reaction, no enantioselectivity was observed and produced only racemic addition products.

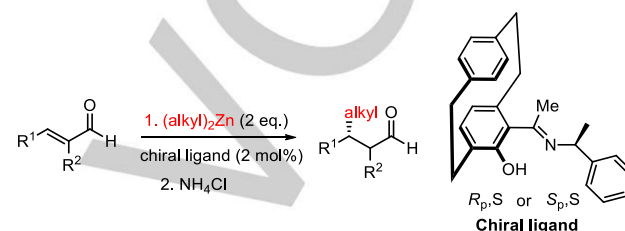


Scheme 20. Asymmetric conjugate addition of dialkylzinc to chalcone and cyclohexenone

3.2. Copper-free asymmetric conjugate addition

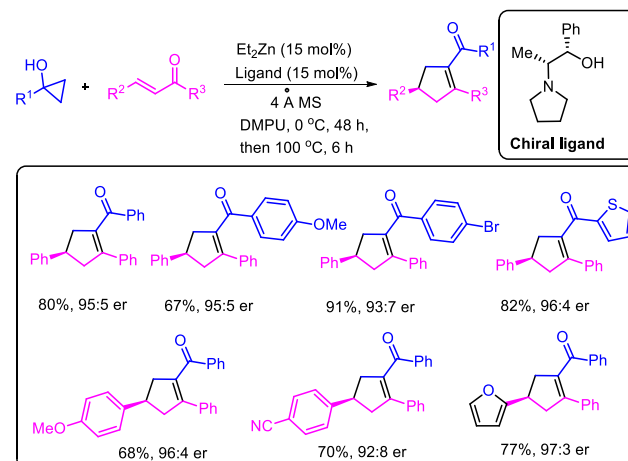
In the asymmetric conjugate addition reactions of organozinc to electron-deficient olefins, copper salts are usually added. The copper cations influence the electronic behaviors of the carbonyl group and therefore, can determine which alkyl/aryl moieties are transferred and also perform as bridging units in the transition state. The first copper-free asymmetric conjugate addition of diethylzinc or diisopropylzinc to α,β -unsaturated aldehydes was

reported by Bräse and Höfener (Scheme 21).^[36] The reactions were carried out in toluene or hexane at -20 °C though the temperature was kept 0 °C for chromone substrate. Though enantioselectivities were excellent, yields were poor because of the formation of little amount of 1,2-addition products (not shown here). Another copper-free asymmetric conjugate addition of dialkylzinc to enones was developed by Cimarelli et al.^[37] Several enantiopure 1-(aminoalkyl)naphthols and 2-(aminoalkyl)phenol were checked as ligands in which Ni(acac)₂ was used as catalyst instead of copper salts. Reactions were performed in CH₃CN solvent at -30 °C. However, in all the cases poor to moderate enantioselectivity were observed (up to 71% ee).



Scheme 21. Asymmetric conjugate addition of dialkylzinc to α,β -unsaturated aldehydes. ^a Ligand (S_p, S) was used whereas ligand (R_p, S) was used in the rest cases.

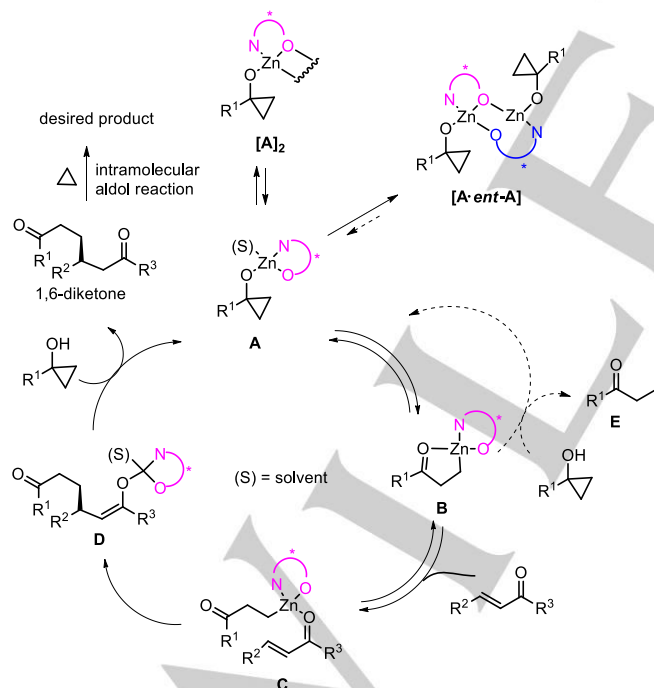
Asymmetric conjugate addition of zinc homoenolate, catalytically generated through the ring opening of a cyclopropanol derivative, to an α,β -unsaturated ketone was recently developed by Yoshikai and Sekiguchi (Scheme 22).^[38]



Scheme 22. Asymmetric conjugate addition of *in situ* generated zinc homoenolate. "er" is enantiomeric ratio.

N,N'-Dimethylpropyleneurea (DMPU) was used as Lewis base and molecular sieve was used as additive. Zinc aminoalkoxide catalyst generated from the reaction of Et_2Zn and a chiral β -amino alcohol promoted the reaction to obtain 1,6-diketones, which further underwent intramolecular aldol condensation upon heating to afford the tri-substituted cyclopentene derivatives with good to high enantioselectivities. Initially the reaction was carried out at 0 °C for 48 h and then heated to 100 °C for 6 h for the subsequent aldol reaction. A variety of 1-substituted cyclopropanols and α,β -unsaturated ketones were successfully used.

A probable mechanism was proposed in which a mixed zinc alkoxide **A** was first formed from the reaction of Et_2Zn , chiral β -amino alcohol and the cyclopropanol. The alkoxide **A** acted as a catalytically competent species in the reaction (Scheme 23). This species then underwent ring opening to form the corresponding homoenolate **B**. This then coordinated to the enone and underwent asymmetric conjugate addition to give a zinc enolate **D**. This enolate then abstracted a proton from the cyclopropanol molecule to give the 1,6-diketone and regenerated the mixed alkoxide **A**. 1,6-Diketone further underwent intramolecular aldol condensation upon heating to produce the desired cyclopentene. If the asymmetric conjugate addition is sluggish the homoenolate **B** can abstract a proton from cyclopropanol to generate the ketone **E**, which can further take part in enolate conjugate addition under zinc catalysis.



Scheme 23. Plausible mechanism for the conjugate addition of *in situ* generated zinc homoenolate

Other catalyst was also used to replace the copper catalyst in the asymmetric conjugate addition of organozinc reagents. For instance, chiral nickel complex catalyzed conjugate addition

of dialkylzinc to enones was reported by Soai et al. in 1988.^[39] This was probably the first report of Ni-catalyzed conjugate addition of organozinc to enones.

Conclusions

The organozinc reagents are highly useful in conjugate addition reactions because of their numerous advantages over other organometallic reagents. For instance, organozinc reagents have broad functional group tolerance, compatibility with transition metals, and simple preparation procedures. Moreover, they prefer 1,4-addition over 1,2-addition unlike many other organometallic reagents thus providing regioselective products. Many exciting results of non-symmetric and asymmetric conjugate additions of organozinc reagents are discussed here, but some of the major contributions are for example, asymmetric conjugate addition of *in situ* generated zinc homoenolate to synthesize tri-substituted cyclopentene (Scheme 22), conjugate addition of *in situ* generated organozinc compound to synthesize pyridine-2-cyclopentane or -cyclohexane (Scheme 9), tandem conjugate addition-cyclization to produce substituted cyclopentane (Scheme 7). In spite of wonderful accomplishments in this area, challenges still remain. For example, in a few reactions (Scheme 7, 14, 18-19) large amount (3 eq.) of organozinc reagents were used and therefore, the reactions became costlier and bulky amount of by-products were also formed. Thus new methods should be developed where low amounts of organozinc may be sufficient for the reaction. In addition, developing environmentally benign and milder conjugate addition reactions of organozinc reagents is highly desirable. Synthetic chemists need a lot of effort and interest to improve the methodology.

Acknowledgements

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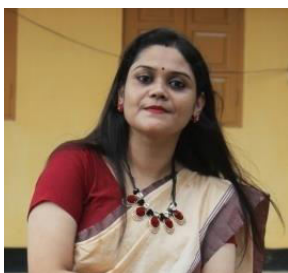
Conflict of Interest

The authors declare no conflict of interest.

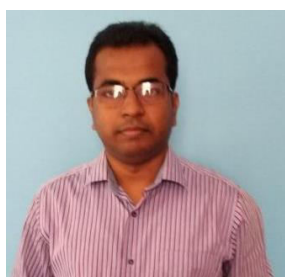
Keywords: Alkylation • Asymmetric catalysis • Conjugate addition • Electron-deficient olefins • Organozinc

- [1] a) E. Erdik, *Organozinc Reagents in Organic Synthesis*, CRC Press, Boca Raton, FL, London, **1996**; b) Z. Rappoport, I. Marek, *The Chemistry of Organozinc Compounds*, John Wiley & Sons, Ltd, Chichester, **2006**; c) P. Knochel, R. D. Singer, *Chem. Rev.* **1993**, 93, 2117–2188; d) A. Deagostino, C. Prandi, S. Tabasso, P. Venturello, *Curr. Org. Chem.* **2011**, 15, 2390–2412.
- [2] a) R. Somanathan, L. Z. Flores-Lopez, R. Montalvo-Gonzalez, D. Chavez, M. Parra-Hake, G. Aguirre, *Min. Rev. Org. Chem.* **2010**, 7, 10-22; b) K. Geurts, S. P. Fletcher, A. W. van Zijl, A. J. Minnaard, B. L.

- Feringa, *Pure Appl. Chem.* **2008**, 80, 5, 1025–1037; c) E. Negishi, *Acc. Chem. Res.* **1982**, 15, 340–348; d) A. A. Zemtsov, A. D. Volodin, V. V. Levin, M. I. Struchkova, A. D. Dilman, *Beilstein J. Org. Chem.* **2015**, 11, 2145–2149.
- [3] A. O. King, N. Okukado, E.-I. Negishi, *J. Chem. Soc., Chem. Commun.* **1977**, 683–684.
- [4] N. Yoshikai, E. Nakamura, *Chem. Rev.* **2012**, 112, 2339–2372.
- [5] a) P. Perlmutter, *Conjugate Addition Reactions in Organic Synthesis*, Pergamon Press, Oxford, **1992**; b) M. E. Jung, *Comprehensive Organic Synthesis*, B. M. Trost, I. Fleming, (Eds.), Vol. 4, Chap. 1.1, Pergamon, Oxford, **1991**, pp. 1–67; c) M. L. Deb, P. J. Bhuyan, *Tetrahedron Lett.* **2007**, 48, 2159–2163; d) M. L. Deb, P. J. Bhuyan, *Synthesis* **2008**, 2891–2898.
- [6] a) B. E. Rossiter, N. M. Swingle, *Chem. Rev.* **1992**, 92, 771–806; b) B. L. Feringa, *Acc. Chem. Res.* **2000**, 33, 346–353; c) M. P. Sibi, S. Manyem, *Tetrahedron* **2000**, 56, 8033–8061; d) N. Krause, A. Hoffmann-Röder, *Synthesis*, **2001**, 171–196.
- [7] A. Alexakis, C. Benhaïm, X. Fournieux, A. van den Heuvel, J.-M. Levéque, S. March, S. Rosset, *Synlett* **1999**, 1811–1813.
- [8] S. J. Degrado, H. Mizutani, A. H. Hoveyda, *J. Am. Chem. Soc.* **2001**, 123, 755–756.
- [9] E. W. Dijk, L. Panella, P. Pinho, R. Naasz, A. Meetsma, A. J. Minnaard, B. L. Feringa, *Tetrahedron* **2004**, 60, 9687–9693.
- [10] L. A. Arnold, R. Naasz, A. J. Minnaard, B. L. Feringa, *J. Org. Chem.* **2002**, 67, 7244–7254.
- [11] a) A. Rimkus, N. Sewald, *Synthesis* **2004**, 135–146; b) T. Jerphagnon, M. G. Pizzuti, A. J. Minnaard, B. L. Feringa, *Chem. Soc. Rev.* **2009**, 38, 1039–1075; c) R. Naasz, R. Imbos, L. A. Arnold, B. Feringa, **2002**, *Copper-catalyzed Enantioselective Conjugate Addition Reactions of Organozinc Reagents*. In *EPRINTS-BOOK-TITLE* University of Groningen, Stratingh Institute for Chemistry.
- [12] a) Y. Guo, S. R. Harutyunyan, *Beilstein J. Org. Chem.* **2020**, 16, 1006–1021; b) D. Pichon, J. Morvan, C. Crévisy, M. Mauduit, *Beilstein J. Org. Chem.* **2020**, 16, 212–232; c) D. Vargová, I. Némethová, R. Šebesta, *Org. Biomol. Chem.* **2020**, 18, 3780–3796; d) J. H. Kim, Y. O. Ko, J. Bouffard, S.-G. Lee, *Chem. Soc. Rev.* **2015**, 44, 2489–2507.
- [13] J. A. Marshall, M. Herold, H. S. Eidam, P. Eidam, *Org. Lett.* **2006**, 8, 5505–5508.
- [14] D. Huang, D.; J.-X. Wang, *Synlett* **2007**, 2272–2276.
- [15] M. D. Aera, F. M. Perna, P. Vitale, A. Altomare, A. Palmieri, L. C. H. Maddock, L. J. Bole, A. R. Kennedy, E. Hevia, V. Capriati, *Chem. Eur. J.* **2020**, 26, 8742–8748.
- [16] N. S. Kondratyev, V. V. Levin, A. A. Zemtsov, M. I. Struchkova, A. D. Dilman, *J. Fluor. Chem.* **2015**, 176, 89–92.
- [17] V. Komanduri, F. Pedraza, M. J. Krische, *Adv. Synth. Catal.* **2008**, 350, 1569–1576.
- [18] S. M. Manolikakes, M. Ellwart, C. I. Stathakis, P. Knochel, *Chem. Eur. J.* **2014**, 20, 12289–12297.
- [19] L. E. Hanna, M. R. Harris, K. Domon, E. R. Jarvo, *Org. Lett.* **2017**, 19, 6304–6307.
- [20] T. J. Fulton, P. L. Alley, H. R. Rensch, A. M. Ackerman, C. B. Berlin, M. R. Krout, *J. Org. Chem.* **2018**, 83, 14723–14732.
- [21] G. Casotti, G. Ciancaleoni, F. Lipparini, C. Nieri, A. Luliano, *Chem. Sci.* **2020**, 11, 257–263.
- [22] A. Alexakis, C. Benhaïm, S. Rosset, M. Humam, *J. Am. Chem. Soc.* **2002**, 124, 5262–5263.
- [23] A. W. Hird, A. H. Hoveyda, *J. Am. Chem. Soc.* **2005**, 127, 14988–14989.
- [24] A. Alexakis, J. E. Bäckvall, N. Krause, O. Pàmies, M. Diéguez, *Chem. Rev.* **2008**, 108, 2796–2823.
- [25] J. Wu, D. M. Mampreian, A. H. Hoveyda, *J. Am. Chem. Soc.* **2005**, 127, 4584–4585.
- [26] E. Fillion, A. Wilsily, *J. Am. Chem. Soc.* **2006**, 128, 2774–2775.
- [27] B.-C. Chen, *Heterocycles* **1991**, 32, 529–597.
- [28] A. Hajra, N. Yoshikai, E. Nakamura, *Org. Lett.* **2006**, 8, 4153–4155.
- [29] K. Endo, M. Ogawa, T. Shibata, *Angew. Chem. Int. Ed.* **2010**, 49, 2410–2413; *Angew. Chem.* **2010**, 122, 2460–2463.
- [30] Q.-L. Zhao, L.-L. Wang, A.-P. Xing, *Tetrahedron: Asymmetry* **2010**, 21, 2993–2998.
- [31] K. Endo, S. Yakeishi, D. Hamada, T. Shibata, *Chem. Lett.* **2013**, 42, 547–549.
- [32] K. Endo, R. Takayama, T. Shibata, *Synlett* **2013**, 24, 1155–1159.
- [33] L. Han, Y. Lei, P. Xing, X.-L. Zhao, B. Jiang, *J. Org. Chem.* **2015**, 80, 3752–3757.
- [34] J. M. García, A. González, B. G. Kardak, J. M. Odriozola, M. Oiarbide, J. Razkin, C. Palomo, *Chem. Eur. J.* **2008**, 14, 8768–8771.
- [35] E. Brüllingen, J.-M. Neudörfl, B. Goldfuss, *New J. Chem.* **2019**, 43, 4787–4799.
- [36] S. Bräse, S. Höfener, *Angew. Chem. Int. Ed.* **2005**, 44, 7879–7881; *Angew. Chem.* **2005**, 117, 8091–8093.
- [37] C. Cimarelli, D. Fratoni, G. Palmieri, *Synth. Commun.* **2009**, 39, 3184–3190.
- [38] Y. Sekiguchi, N. Yoshikai, *J. Am. Chem. Soc.* **2021**, 143, 4775–4781.
- [39] K. Soai, S. Yokoyama, T. Hayasaka, K. Ebihara, *J. Org. Chem.* **1988**, 53, 4148–4149.

Author's biography

Dr. Biswajita Baruah was born in Jorhat, Assam, India. She completed her BSc. from J. B. College, Jorhat and MSc. from Dibrugarh University, Assam. She cleared National Eligibility Test (NET, LS) and joined Dr. Bhuyan's research group at North East Institute of Science and Technology (NEIST), Jorhat. After obtaining her PhD degree in 2010, she joined the Department of Chemistry, Pandu College in 2011 as assistant professor and has been working till date. Her current research interest includes development of novel catalyst-free synthetic methodologies.



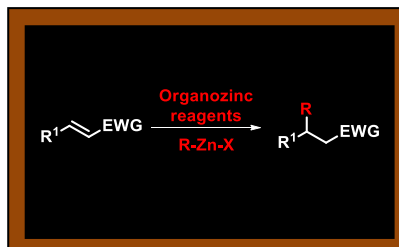
Dr. Mohit. L. Deb was born in Tripura, India. He completed BSc., MSc. from Tripura University and obtained PhD degree (2008) under the supervision of Dr. P. J. Bhuyan at North East Institute of Science and technology (NEIST)-Jorhat, Assam. He then joined Dr. K. R. Prasad group (2009) at Indian Institute of Science (IISc) as a Research Associate where he worked on asymmetric catalysis. He worked in Jubilant Chemsys Ltd, India (2010-12) as a Research Scientist. He completed his postdoctoral research on C-H functionalizations from ITQB-UNL, Lisbon with Prof. Christopher D. Maycock. In 2014, he joined the Department of Applied Sciences, Gauhati University as a guest faculty and later awarded the DST-Young Scientist position. His current research area is sustainable C-H functionalizations and organometallic chemistry. He has around 50 publications in reputed international journals.

Entry for the Table of Contents

Layout 1:

MINIREVIEW

This review discusses the efficient reactions related to alkylation of electron-deficient olefins through the conjugate addition of organozinc reagents, including their advantages, limitations and possibilities for improvement. Both asymmetric and non-asymmetric conjugate additions are highlighted. Cyclic and acyclic enones, enals, β -nitrostyrenes have been used as electron-deficient olefins.



Organometallic Chemistry

Biswajita Baruah,^[a] Mohit L. Deb,^{*,[b]}

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Alkylation of electron-deficient olefins through conjugate addition of organozinc reagents: An update